



Kinematics: Constant Acceleration

• Worksheet 1

Mechanics Chapter 2: Kinematics: Constant Acceleration. Answer all questions. Show your working clearly. Total: 81 marks. Time: 100 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are four sections, one for each topic in Chapter 2. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Take $g = 9.8 \text{ m s}^{-2}$. State your positive direction, and give answers to 3 significant figures.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Displacement, velocity, speed and acceleration	16	
B The suvat equations	20	
C Displacement–time and velocity–time graphs	20	
D Vertical motion under gravity	20	
Roar-level challenge	5	
Total	81	

Section A Displacement, velocity, speed and acceleration

16 marks

1. A car accelerates uniformly from 5 m s^{-1} to 25 m s^{-1} in 8 s. Find its acceleration. [2]

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2. A cyclist slows uniformly from 20 m s^{-1} to 8 m s^{-1} in 4 s. Find her acceleration, and state her deceleration. [2]

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3. A cyclist rides 400 m due east in 50 s, then 200 m due west in 30 s. [3]
Find (a) the average speed, (b) the final displacement from the start, (c) the average velocity.

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4. A train accelerates uniformly from rest to 30ms^{-1} in 60 s. [3]
(a) Find the acceleration of the train.
(b) Find the time taken for the train to reach 18ms^{-1} .
(c) Explain the difference between the train's speed and its velocity.

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5. A particle passes a point A with speed 4 m s^{-1} . It accelerates uniformly at 0.5 m s^{-2} for 10 s, and then decelerates uniformly to rest in a further 6 s.

(a) Find the speed of the particle after 10 s. [2]

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(b) Find the deceleration in the second stage. [1]

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(c) Find the average speed of the particle over the 16 s. [3]

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(Total for Question 5 is 6 marks)



Section B The suvat equations

20 marks

6. A particle has initial velocity 3 m s^{-1} and constant acceleration 2 m s^{-2} . Find its velocity after 5 s. [2]

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7. A car starts from rest and accelerates uniformly at 1.5 m s^{-2} over 48 m. [3]
Find its final speed and the time taken.

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8. A car travelling at 20 m s^{-1} brakes uniformly and stops in 50 m. [3]
Find the deceleration and the time taken to stop.

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9. A particle passes a point A with speed 6 m s^{-1} , moving with constant acceleration 2 m s^{-2} . It passes a point B 8 s later. [4]

(a) Find the distance AB and the speed at B .

(b) Find the speed of the particle at the midpoint of AB .

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10. A car P passes a point O on a straight road, moving at a constant speed of 15 ms^{-1} . At the same instant, a car Q starts from rest at O and moves in the same direction with constant acceleration 2.5 ms^{-2} .

(a) Find the time taken for Q to catch up with P . [3]

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(b) Find the distance from O at which Q catches up with P . [1]

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(c) Find the speed of Q at this moment. [1]

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(d) Find the greatest distance between the cars before Q catches up with P . [3]

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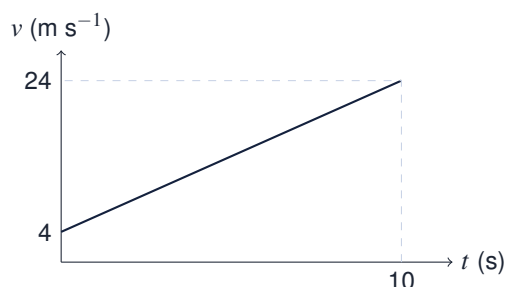
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(Total for Question 10 is 8 marks)

**Section C Displacement–time and velocity–time graphs**

20 marks

11. The velocity–time graph shows the motion of a particle for 10 s.

[2]

Find (a) the acceleration, (b) the distance travelled.

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12. A particle accelerates uniformly from rest to 12 m s^{-1} in 4 s, travels at 12 m s^{-1} for 10 s, then decelerates uniformly to rest in 6 s.

[3]

- (a) Sketch a velocity–time graph of the motion.
- (b) Find the total distance travelled.
- (c) Find the deceleration.

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13. A car starts from rest and accelerates at 3 ms^{-2} for 5 s. It then travels at constant speed for 20 s, [3]
and then decelerates uniformly to rest in 10 s.
Find the total distance travelled by the car.

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14. A train starts from rest and accelerates uniformly for T seconds to 20 ms^{-1} . It travels at 20 ms^{-1} for [4]
60 s, then decelerates uniformly to rest in 20 s. The total distance travelled is 1900 m.
Find T , and the acceleration in the first stage.

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15. A car starts from rest and accelerates uniformly to a speed of $V \text{ ms}^{-1}$ in 10 s. It then travels at $V \text{ ms}^{-1}$ for 30 s, and then decelerates uniformly to rest in 15 s. The total distance travelled is 1275 m.

(a) Sketch a velocity–time graph for the motion. [2]

(b) Show that $V = 30$. [3]

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(c) Find the total time for which the car is travelling at more than 20 ms^{-1} . [3]

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(Total for Question 15 is 8 marks)



Section D Vertical motion under gravity

20 marks

16. A ball is dropped from rest from a height of 20 m. Find the time it takes to reach the ground. [2]

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17. A ball is thrown vertically upwards at 14.7 ms^{-1} . Find the greatest height it reaches and the time taken to reach it. [3]

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18. A stone is thrown vertically downwards at 5 ms^{-1} from the top of a building 30 m high. Find the speed of the stone as it hits the ground and the time taken. [3]

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19. A ball is projected vertically upwards at 20 ms^{-1} from a point O . Find the length of time for which the ball is at least 15 m above O . [4]

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20. A stone is projected vertically upwards at 21 m s^{-1} from a point A, which is 28 m above the ground. The stone moves freely under gravity until it hits the ground.

(a) Find the greatest height of the stone above the ground. [2]

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(b) Find the time taken for the stone to hit the ground. [3]

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(c) Find the speed of the stone as it hits the ground. [2]

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(d) State one limitation of the model. [1]

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(Total for Question 20 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

21. A ball A is dropped from rest from a point 45 m above the ground. At the same instant, a ball B is thrown vertically upwards at 21 m s^{-1} from the ground directly below. Find the time at which the balls meet, their height above the ground at that moment, and the velocity of B at that moment. [5]

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END OF WORKSHEET

Total: 81 marks